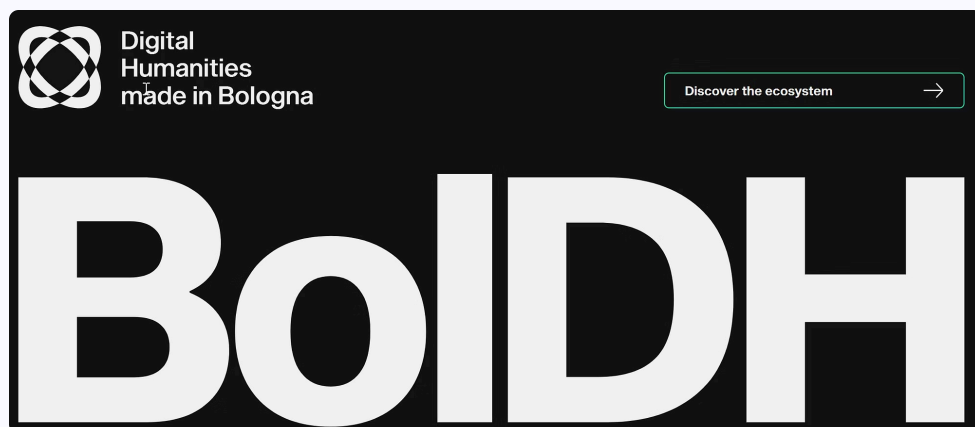




What Do We Mean by Scholarly Digital Culture?

VENICE SUMMER SCHOOL IN DIGITAL AND PUBLIC HUMANITIES

Ca' Foscari University · Ca' Dolfin · VeDPH · Venice State Archive

July 6th, 2026



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BolDH – The Bologna Digital Humanities Ecosystem

Education · Research · Projects · Infrastructures · Dissemination... Digital products!

Over the past decade, the University of Bologna has built a comprehensive ecosystem for Digital Humanities – from graduate education to research infrastructure – united under the **BolDH** umbrella.

DHDK · 2017

International Master's degree in *Digital Humanities and Digital Knowledge*. A two-year program training the next generation of humanists.

DH.arc · 2018

The *Digital Humanities Advanced Research Center* – Bologna's hub for interdisciplinary DH research, methodology, and collaboration.

ADLab · 2019

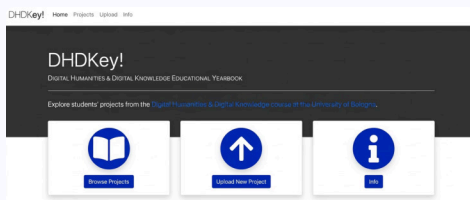
Laboratory for digital acquisition of heterogeneous materials – from manuscripts to physical artifacts – supporting digitization workflows.

DL FICLIT · 2019

The *Digital Library* for storing and publishing semantic data, housing scholarly digital objects and structured collections.

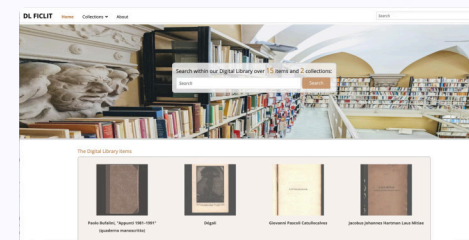
CHeDE · 2022

PhD program on *Cultural Heritage in the Digital Ecosystem* – Bologna's flagship doctoral offering in digital cultural heritage.



DHDKey! – Digital Humanities Educational Yearbook

At every moment of the BolDH ecosystem, complex digital objects are produced: each component – DHDK, DH.arc, ADLab, DL FICLIT, and CHeDE – generates **digital artifacts** as part of its core activity.



DL FICLIT – Bologna's Semantic Digital Library

A Necessary Premise: The Expanding Notion of Scholarly Output

In digital humanities research, we do not produce data alone: we also create the **applications, tools, and interfaces** needed to interpret, navigate, and make sense of that data. These outputs are inseparable, and together they mark a profound **epistemological transformation** in what counts as knowledge production.

Digital Outputs Across the Project Lifecycle

- **Creation**

Raw data, datasets → data entry tools, acquisition systems

- **Description**

Metadata, annotations → annotation platforms, ontologies

- **Management**

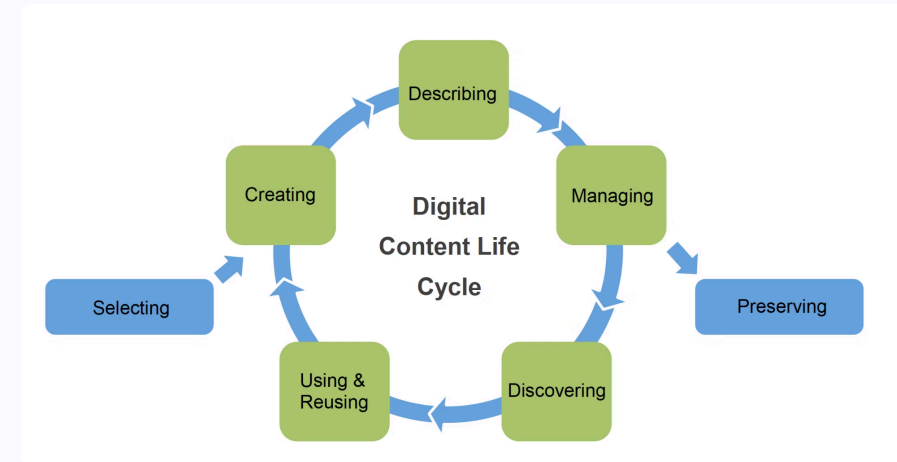
Repositories, versioned archives → APIs, data management systems

- **Discovery**

Indexes, knowledge graphs → search interfaces, visualization apps

- **Use**

Published results → web applications, interactive editions



- At every stage of the project lifecycle – from creation to use – we produce data and applications that give that data meaning. We create **digital research products**

- We will return to the question of **preservation** later.

What do we mean by Scholarly Digital Products?

What does a **scholarly digital product** actually look like? Across humanities disciplines, DH research now produces a spectrum of digital artifacts – from richly annotated datasets and semantic knowledge graphs, to interactive web applications, digital critical editions, and ontology-based models.



Knowledge Graphs

Semantic networks connecting people, places, concepts, and institutions in structured, queryable form



Ontologies and controlled vocabularies

Formal conceptual models that define domain vocabularies and relationships according to shared standards



Web Applications

Interactive platforms for exploring, browsing, and contributing to digital collections and research outputs



Tools & Software

Scripts, pipelines, computational notebooks, and processing tools that transform, analyze, and operationalize research data

Rethinking Scholarly Digital Products as part of the Cultural Heritage

The KNOT Project, 2023–2025

[ICDP-DIGITAL-LIBRARY.GITHUB.IO/KNOT/](https://icdp-digital-library.github.io/knot/)

KNOT – is a three-year pilot project from the University of Bologna and the Central Institute for the Digitization of Cultural Heritage (ICDP), funded by the Digital Library of the Italian Ministry of Culture.

The project investigates how to **describe**, **valorize**, and **make visible** the scholarly digital objects (SDOs) produced by humanities research situating them explicitly within the national cultural heritage framework.

KNOT - Rethinking Scholarly Digital Objects as Cultural Heritage ENG IT

Supporting a New Digital Ecosystem of Culture...

KNOT is a three-year pilot project from the University of Bologna and the Central Institute for the Digitization of Cultural Heritage (ICDP) - Digital Library of the Italian Ministry of Culture to showcase the Digital Cultural Heritage of Italian universities.

The project is part of an agreement between the University of Bologna and the Ministry to establish a research infrastructure to support the digital ecosystem of culture that is a central part of the [ICDP's mission](#). To this end the University of Bologna is carrying out parallel and connected research initiatives in three departments (Department of Cultural Heritage - DBC, Department of Classical Philology and Italian Studies - FICLIT, and Department of Computer Science and Engineering - DISI) with the goal of investigating ways to integrate the Digital Cultural Heritage (DCH) of Italian universities within the national infrastructure.

KNOT is undertaken within the [Digital Humanities Advanced Research Center \(/DHARC\)](#), part of the FICLIT department, with support from DISI, and in collaboration with the DBC, which is undertaking a two-year census (2022-2024) of the digital and digitized cultural heritage of Italian universities.

... By Rethinking Scholarly Digital Objects as Cultural Heritage

For years now academic research has operated in a primarily digital space, with digital tools playing an increasingly important role in both the activity of research and the production of results as digital-born and/or digitized objects. As a result of this, universities find themselves already in possession of a multifaceted and interesting, and so far unexplored, example of DCH in the form of digital objects created by academic research projects.

These scholarly digital objects are the focus of investigation of the KNOT project with the goal to explore ways to describe and valorize them within the context of [IPoC, the national infrastructure being built by the ICDP](#). Our investigation starts from the field of the Humanities, and in particular the Digital Humanities (DH), where a project-oriented and multidisciplinary approach often results in the creation of digital objects that recontextualise existing cultural heritage — such as digital editions, corpora, visualization tools, collections — and thus offer a particularly interesting example of the scholarly digital object as DCH.

From this conceptual base, KNOT leverages the Digital Library's infrastructure alongside practices from the field of DH to create a web application and guidelines for the collection, management, enrichment, and reuse of these digital scholarly objects. In doing so, the project will also confront some key issues facing these objects including their visibility within the Italian academic system and the lack of a shared framework for their display and preservation, which affects their documentation and classification among other things.

The development of a national infrastructure for DCH such as the one currently being undertaken by the ICDP offers a unique opportunity for Italian universities to engage with this segment of their DCH holdings in new ways. By focusing on the scholarly digital objects produced by humanities research, and in particular DH, the KNOT project seeks to provide a useful base from which to let new insights emerge around how to valorize these objects both within and outside of academia.

The Pilot

Learn more about the three-year plan, research lines, and outputs.

The Data Model

Learn more about the conceptual base of the project, its structure and components.

The Ontology

Learn more about the KNOT Ontology, which expresses the data model in RDF.

Three Pillars

The Pilot

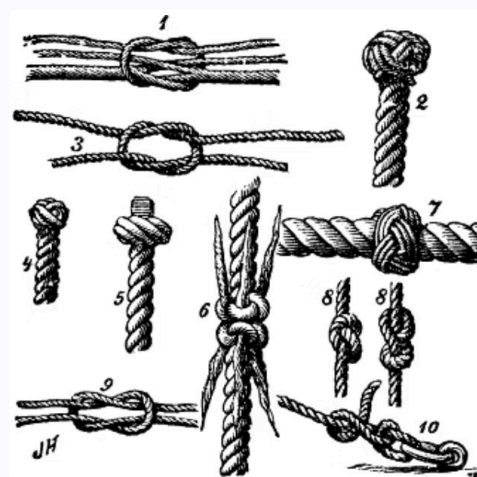
Three-year research lines, labs, and outputs

The Data Model

Conceptual base, project structure, and components

The Ontology

RDF data model for describing SDOs



PROJECTS.DHARC.UNIBO.IT/KNOT/

[illegible]

Category	Percentage
Database	24
Digital Edition	15
Digital Platform	13
Digital Library	9
Digital Catalogue	6
Digital Archive	5
Storytelling Platform	5
Endpoint	3
Research Platform	3
Software	3
Corpus	2
Crowdsourcing Platform	2
Digital Repository	2
Knowledge Base	2
Image Database	1
Semantic Digital Edition	1

Field	Number of Publications (approx.)
Informatics	44
Archival science, bibliography and librarianship	33
Italian studies and comparative literature	17
Medieval latin and romance philologies and literatures	14
Archaeological sciences	13
Historical and general linguistics	13
Musicology and history of music	8
Italian linguistics and philology	7
Libretto	6
Art History	5
Dante Alighieri	4
Opera	4
Modern History	4
Latin Epigraphy	4

What are the research projects and their objects about?

KNOT – An entry in CLEF: The Zeri-LODE Project

PROJECTS.DHARC.UNIBO.IT/KNOT/VIEW-ZERI-LODE-PROJECT

KNOT Browse SPARQL endpoint Visualization

Zeri&LODE

DESCRIPTION
Zeri & LODE is a project to present cataloguing data belonging to art historical photo Photography domain (DAEntry Ontology, Fentry Ontology and HICO Ontology) and pr

PRIMARY SUBJECTS
Archival science, bibliography and librarianship, Informatics, Art history, Federico Zeri

ASSOCIATED ENTITIES
Fondazione Federico Zeri, Università di Bologna, Digital Humanities Advanced Research Centre (UDH.Arc).

LOCATION
Bologna

TECHNOLOGIES USED
RDF, XML, SPARQL Ontologies, CIDOC CRM, SPARQL, XSLT.

BIBLIOGRAPHIC REFERENCES
<http://dx.doi.org/10.1109/JCDL.2014.6970182>

PROJECT STATUS
Completed

Abstract
This paper presents the first steps of a project to convert the notable Italian "Zeri photo archive" (Zeri photo archive) into a Linked Open Data (LOD) dataset. The full project entails the analysis of the records' description model (Schema F) in exploring existing data models, the creation of the RDF triple store, the creation of a user interface for browsing the linked open dataset. This paper presents and discusses the first steps of the project.

Published in: IEEE/ACM Joint Conference on Digital Libraries
Date of Conference: 08-12 September 2014
DOI: 10.1109/JCDL.2014.6970182

Contributors: Fabio Vitali, Mariellen Daquino, Francesca Martini, Francesca Tomasi

Research Activities: Modeling, Disseminating, Linked Open Data.

Outputs: Zeri Photo Archive RDF Dataset.

Home Page: <https://data.fondazionezeri.unibo.it/>

Art history

Research Project
This record catalogues scholarly digital activity as an instance of the PROV-O Activity class. This activity represents academic research projects that have produced digital scholarly objects. See the Documentation page for more information.

PERMALINK
<http://w3id.org/knot/data/zeri-lode-project>

EXTERNAL CONCEPT
This page implements an external concept (from the TaDiRAH taxonomy) and links it to the project's internal concept (from the project's internal ontology).

TaDiRAH: Taxonomy of Digital Research Activities in the Humanities
Vocabularies About Editor API
Interface language: English

Abstract
This paper presents the first steps of a project to convert the notable Italian "Zeri photo archive" (Zeri photo archive) into a Linked Open Data (LOD) dataset. The full project entails the analysis of the records' description model (Schema F) in exploring existing data models, the creation of the RDF triple store, the creation of a user interface for browsing the linked open dataset. This paper presents and discusses the first steps of the project.

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People → Entity → VIAF

Authority control for persons linked to the Virtual International Authority File

Places → Entity → GeoNames

Authority control for locations linked to the GeoNames geographic authority record

Bibliographic reference → DOI → Paper

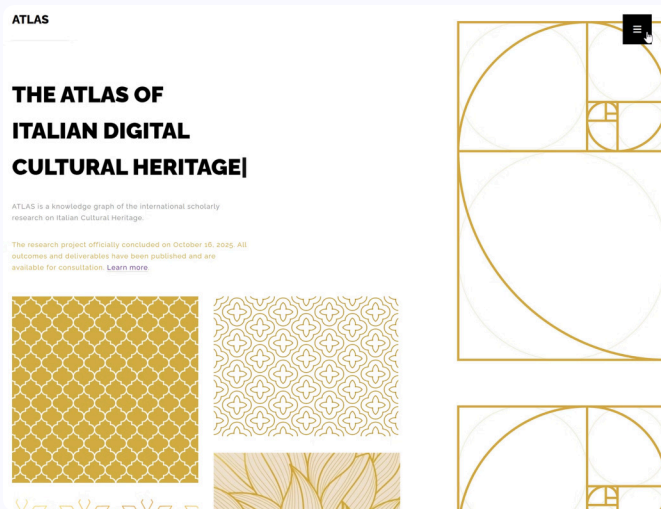
Publications linked via DOI to the paper online

Activity → Taxonomy → TaDiRAH

Research activities linked to the TaDiRAH taxonomy of digital research activities in the humanities

Subject → Wikidata → Other KNOT projects

Research subjects linked to all other KNOT projects sharing the same topic



[DH-ATLAS.GITHUB.IO/](https://github.com/DH-ATLAS)

The PRIN ATLAS Project

The **ATLAS** project builds a national Knowledge Graph (KG) of Italian Digital Humanities research.

It maps the DH landscape by encoding semantic relationships across the discipline: **people and organizations, places, dates,** and multiple levels of the notion of *concept* – including resource types, academic fields, subjects, and technologies.

Discover new (hidden) knowledge

Aim

Connecting researchers, institutions, projects, and outputs it makes the **invisible** infrastructure of humanities research **findable, citable, and reusable**.

Our classification of scholarly digital products/projects/objects

Digital Scholarly Editions

Linked Open Data

Ontologies

Software

Text Collections

General Recommendations

ATLAS Outcomes and the role of Guidelines for FAIRification

DH-ATLAS.GITHUB.IO/GUIDELINES/

For a digital product to be truly **scholarly**, it must follow **guidelines and best practices** that guarantee its FAIRness – making it Findable, Accessible, Interoperable, and Reusable. ATLAS has produced a comprehensive suite of outputs serving this goal.

Web Application

Interactive portal for browsing the ATLAS KG

Ontology

Formal data model for DH scholarly objects

Knowledge Graph

Populated semantic graph of Italian DH research

CLEF

Classification and labeling framework

White Book + Guidelines

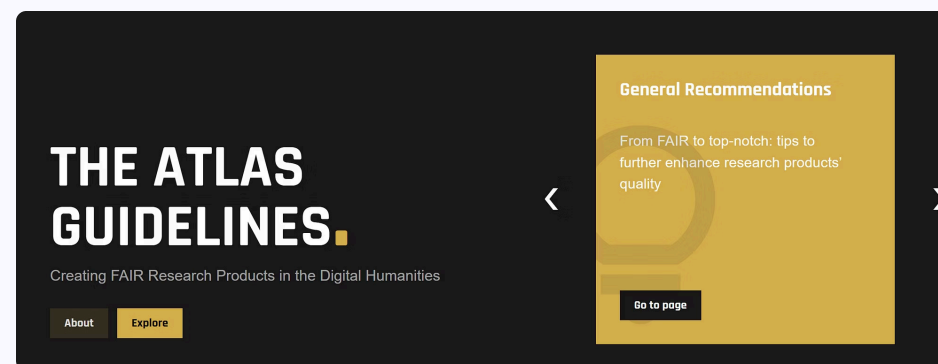
Policy and best-practice documentation

ETV + REST API + OAI-PMH

Vocabularies and interoperability services

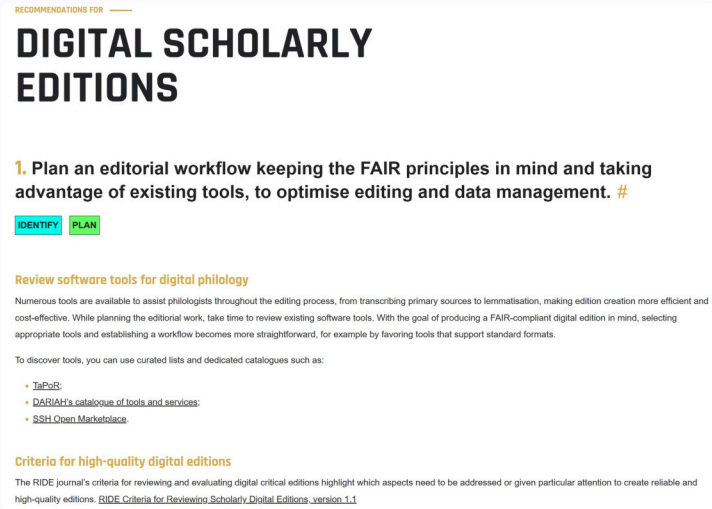
We understood that guidelines are needed!

These **cross-cutting recommendations** focus on practical steps that **make any research product easier to find, cite, and use** – from clear landing pages with persistent identifiers to English metadata and multi-format documentation.



The ATLAS (FAIR) Guidelines – Structure and Best Practices

The ATLAS Guidelines are themselves **designed according to the same principles they advocate**. Each guideline entry is structured around a consistent schema that ensures clarity, reproducibility, and practical applicability for any digital research workflow.

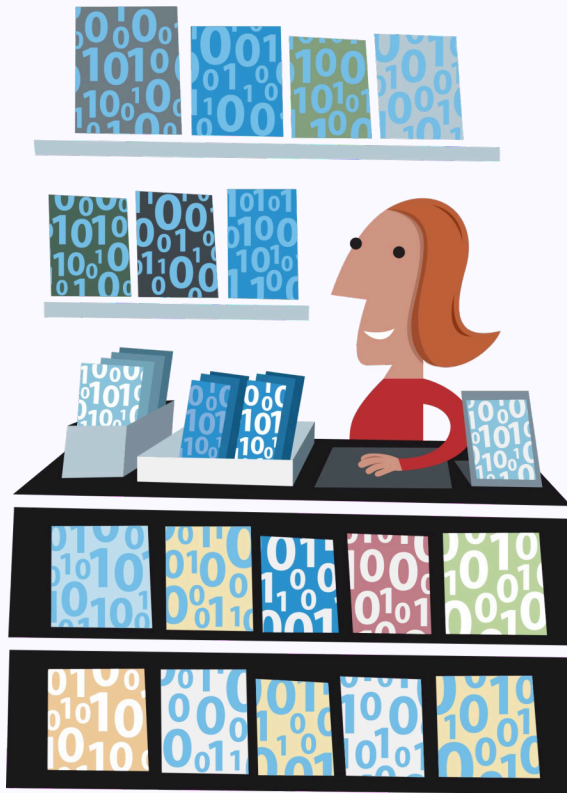


Each entry includes:

- **Type**
Category of the digital object
- **Step**
The single moment of the workflow for that type
- **Activity/ies**
Where in the project lifecycle it/them apply/ies
- **Description**
Rationale and contextual explanation
- **Examples + References**
Illustrative cases and further reading where available

📄 Even guidelines must follow good practices – meta-documentation as a scholarly standard.

- **IDENTIFY** - recommendations focused on identifying your own research data, community standards, best practices, and existing resources such as tools and datasets that may be useful;
- **PLAN** - recommendations about planning your research product lifecycle and what to consider from creation through publication and maintenance;
- **PRODUCE** - recommendations for creating and formatting your research product, with particular attention to formats, metadata, and data models;
- **DEPOSIT** - recommendations for properly depositing and preserving your research data, including considerations for licences and legal aspects;
- **DISSEMINATE** - recommendations on how to effectively reach your audience and make your data (re-)usable.



www.digitalbevaring.dk

If Scholarly Digital Objects are part of the Cultural Heritage they need to be preserved.

What Preservation Really Means

To preserve a digital research product means maintaining over time **both access to the data** and to the **tools and environments** used to create, manipulate, and manage that data. This includes software dependencies, APIs, rendering environments, and computational contexts – not just archived files.

Physical, logical and conceptual preservation.

The Core Challenge

- Strategies
- Funding
- Obsolescence
- Institutions

And how about preservation?

Guidelines address best practices for every stage of a digital project's lifecycle – but they only *partially* resolve the deeper challenge of **long-term sustainability**. Preservation is not merely a technical problem; it is a **structural and institutional one**.

Main Difficulties in Preserving Digital Scholarly Products

Despite growing awareness of the importance of digital preservation, humanities research projects face a cluster of interrelated challenges that make long-term sustainability genuinely difficult to achieve in practice.



Technological Obsolescence

Software libraries, frameworks, and infrastructure evolve rapidly, making it difficult to maintain the computational environments needed to run or interpret older digital products.



Funding Discontinuity

Project-based funding structures rarely support long-term maintenance. When grants end, so does the active stewardship of the digital outputs produced.



Insufficient Documentation

Many projects document their final outputs but not their workflows, design choices, or dependencies – making future reuse and migration extremely difficult.



Lack of Institutional Ownership

It is often unclear which institutional actor – department, library, research center, or ministry – bears responsibility for preserving a given digital scholarly object.

The Role of Institutions

Ensuring the long-term survival of digital scholarly products requires more than individual researcher effort – it demands a coordinated **institutional response**. Universities, national libraries, cultural heritage ministries, and infrastructure providers each have a part to play in building a sustainable digital scholarly ecosystem.

Universities & Research Centers

Establish internal policies, repositories, and support structures for DH outputs. Embed preservation planning in research workflows from project inception.

Libraries & Archives

Extend their traditional custodial role into the digital domain – acquiring, cataloguing, and ensuring long-term access to scholarly digital objects alongside physical collections.

Funding Bodies & Ministries

Mandate preservation planning as a condition of funding; support dedicated national infrastructure for digital cultural heritage beyond individual project timelines.

International Networks

Initiatives like CLARIN, DARIAH, and EOSC offer shared infrastructure and standards, reducing duplication and enabling interoperability across national boundaries.

Open Questions

Even as frameworks, guidelines, and infrastructures improve, a number of **fundamental questions** remain open questions that sit at the intersection of **epistemology, policy, and practice** for anyone working in digital humanities today.

What counts as a scholarly contribution?

How do we evaluate datasets, ontologies, or software tools as primary research contributions in academic review, hiring, and promotion processes?

How do we ensure FAIRness in practice?

FAIR principles are widely endorsed but unevenly implemented. What technical and cultural changes are needed to close the gap between principle and practice?

Who is responsible for preservation?

In a landscape of project-based funding and distributed infrastructure, how do we assign clear, durable institutional responsibility for digital scholarly objects?

How do we train the next generation?

What skills, literacies, and professional identities do early-career humanists need to thrive in a digital scholarly ecosystem – and how should curricula respond?



Pròdigi – A Data Journal

PRODIGHI.UNIBO.IT

«**Pròdigi** (digital products) is the international venue for the collection and valorisation of **research outputs in the humanities** that emerge from digital publications, understood not merely as dissemination tools but as **research objects**. The journal welcomes peer-reviewed scholarly contributions that illustrate the **scholarly value, research processes, and reusability potential** of the digital products presented.»

Pròdigi addresses a critical gap in the scholarly communication landscape: the absence of a peer-review venue dedicated specifically to digital products as first-class academic contributions. By publishing articles that critically describe and assess digital outputs, the journal helps establish norms, build a community of practice, and make DH research outputs **visible, citable, and evaluable** within the broader humanities ecosystem.

Digital Scholarly Culture

A Theoretical and Physical Layered Infrastructure

Understanding scholarly digital culture means recognizing that it is a **stack**, encompassing people, data, technology, and process in an integrated system.



Thank You!

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Projects Referenced

- [ADLab](#)
- [ATLAS Guidelines](#)
- [BolDH](#)
- [CHeDE](#)
- [DH.arc Research Center](#)
- [DHDK](#)
- [DHDKKey](#)
- [DL FICLIT](#)
- [KNOT Project](#)
- [KNOT Web Application](#)
- [PRIN ATLAS](#)
- [Pròdigi Journal](#)